

## Quiz 5 — Energy, Laziness, and a Coin: Solutions

M 375T Fall 2026 — optional quiz, 10 points. Q1: 4 pts; Q2: 4 pts; Q3: 2 pts.

1. (a)  $d = (3, 3, 2, 2)$  and

$$L = D - W = \begin{pmatrix} 3 & -2 & 0 & -1 \\ -2 & 3 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

*1 pt:  $d$  on the diagonal,  $-W$  off it; rows must sum to 0.*

(b) Only edges where  $v$  changes sign contribute:  $1-2$  (weight 1, jump 2) and  $3-0$  (weight 1, jump 2), each  $1 \cdot 2^2 = 4$ . Line 1 prints 8.0. The edges  $0-1$  and  $2-3$  have  $v_i = v_j$  and contribute 0. *1 pt: the number and which edges.*

(c)  $(Lv)_i = d_i v_i - \sum_j W_{ij} v_j$ : vertex 0:  $3 - (2 - 1) = 2$ ; vertex 1:  $3 - (2 - 1) = 2$ ; vertex 2:  $-2 - (1 - 1) = -2$ ; vertex 3:  $-2 - (1 - 1) = -2$ . Line 2 prints  $[2, 2, -2, -2] = 2v$ , so  $v$  is an eigenvector of  $L$  with eigenvalue 2. (Consistent with (b):  $v^T L v = 2 v^T v = 2 \cdot 4 = 8$ .) *1 pt: the vector and the eigenvalue; “eigenvector” without the check earns  $\frac{1}{2}$ .*

(d)  $\text{cut}(A, B) = W_{12} + W_{30} = 1 + 1 = 2$ ,  $\text{vol}(A) = d_0 + d_1 = 6$ ,  $\text{vol}(B) = d_2 + d_3 = 4$ ,  $\text{Ncut} = 2(\frac{1}{6} + \frac{1}{4}) = \frac{5}{6} \approx 0.83$ . Relation: for  $v = \mathbf{1}_A - \mathbf{1}_B$  every cut edge has jump 2, so  $v^T L v = 4 \text{cut}(A, B) = 8 = 4 \cdot 2$ . (Equivalently  $\mathbf{1}_A^T L \mathbf{1}_A = \text{cut}(A, B) = 2$ .) *1 pt: cut, volumes, Ncut, and the factor 4 (or the indicator identity). [4 pts]*

2. (a)  $W = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ ,  $d = (2, 3, 1)$ ,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 1 & 0 \end{pmatrix}, \quad \pi = \frac{d}{\sum_i d_i} = \left(\frac{1}{3}, \frac{1}{2}, \frac{1}{6}\right).$$

*1 pt:  $\frac{1}{2}$  for  $P$  (rows sum to 1),  $\frac{1}{2}$  for  $\pi$  normalized.*

(b)  $\mu_1 = P^T \mu_0 = \text{row 1 of } P = (\frac{2}{3}, 0, \frac{1}{3})$ ;  $\mu_2 = P^T \mu_1 = \frac{2}{3} \cdot \text{row 0} + \frac{1}{3} \cdot \text{row 2} = (0, 1, 0) = \mu_0$ . The law alternates  $\mu_0, \mu_1, \mu_0, \mu_1, \dots$  forever and never converges. Reason: the graph is bipartite,  $\{1\}$  against  $\{0, 2\}$ ; every step changes side. The responsible eigenvalue of  $P$  is  $-1$ : the eigenvalues of  $P$  are  $1, 0, -1$ , and  $v = (1, -1, 1)^T$  (+1 on one side,  $-1$  on the other) satisfies  $Pv = (-1, 1, -1)^T = -v$ . Connectedness gives a unique  $\pi$ , not convergence. *1.5 pts:  $\frac{1}{2}$  for  $\mu_1, \mu_2$ ;  $\frac{1}{2}$  for “alternates, bipartite”;  $\frac{1}{2}$  for the eigenvalue  $-1$ .*

(c)  $Q = \frac{1}{2}(I + P) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ . Lazy law after one step from vertex 1:  $Q^T e_1 = \text{row 1 of } Q$

$Q = (\frac{1}{3}, \frac{1}{2}, \frac{1}{6}) = \pi$  exactly. After two steps (and forever after) it stays  $\pi$ , because  $Q^T \pi = \pi$ . *1 pt:  $\frac{1}{2}$  for the row,  $\frac{1}{2}$  for noticing it equals  $\pi$  and therefore stays.*

(d) Need (i)  $|\eta_k| < 1$  for all  $k \geq 2$ , i.e. the eigenvalue 1 is simple: this comes from **connectedness**; (ii)  $\eta_k \geq 0$ , i.e. no eigenvalue  $-1$ : this comes from **laziness**,  $\eta = \frac{1}{2}(1 + \lambda)$  maps  $[-1, 1]$  to  $[0, 1]$  and sends  $-1$  to 0. Then every term  $a_k \eta_k^t w_k \rightarrow 0$ . (Here  $\eta = (1, \frac{1}{2}, 0)$ : the  $-1$  mode of (b)

became 0 and dies in one step, which is why (c) hit  $\pi$  immediately.)  $\frac{1}{2}$  pt: both facts with the right source each. [4 pts]

3. (a) (20000,): one count per row. `k.mean()`  $\approx 800 = 1600 \cdot \frac{1}{2}$ . `k.std()`  $\approx \sqrt{1600 \cdot \frac{1}{2} \cdot \frac{1}{2}} = 20$  (Binomial(1600,  $\frac{1}{2}$ )). 1 pt:  $\frac{1}{3}$  each; the shape must be a tuple with one entry.
- (b)  $\varepsilon = \frac{850}{1600} - \frac{1}{2} = 0.03125$ ,  $D \approx 2\varepsilon^2 = 0.00195$ ,  $2nD \approx 2 \cdot 1600 \cdot 0.00195 = 6.25 > 3.84$ : rejected. (Exact  $D = 0.001954$ ,  $2nD = 6.254$ .) Check:  $|k - 800| = 50 > 1.96 \cdot 20 = 39.2$ , same decision. 1 pt:  $\frac{1}{2}$  for  $2nD$  and the decision,  $\frac{1}{2}$  for the standardized check. [2 pts]

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Q1 mirrors practice F1(b)–(c),(e) (Dirichlet form,  $Lv$ , cut/vol/Ncut) on the Quiz 4 graph; Q2 mirrors D2, E1 and F3 (weighted path, invariant measure, lazy walk, why the lazy walk converges) with the bipartite oscillation of Lecture 8 §3 / Lecture 10; Q3 mirrors C2 and Practice 4 A1–A3, B1 (standardize a count,  $D \approx 2\varepsilon^2$ ,  $2nD > 3.84$ ) with new numbers.